Aether

Data Analysis Report

Empirical diagnostic evaluation, descriptive statistics, and multivariate exploratory analysis.

DATASET NAME	**customer_metrics.csv**
REPORT DATE	07 October 2026, 20:24 UTC
DATASET VERSION	Version 2
TOTAL RECORDS	20 rows
FEATURE DIMENSIONS	4 columns (3 numeric, 1 categorical)
DATA QUALITY SCORE	**100/100**
SCOPE OF ANALYSIS	Full-Spectrum Exploratory & Statistical Audit

Notice: This report is generated dynamically from verified local dataset records. All numerical metrics, distribution profiles, and correlation coefficients are computed strictly from the active dataset version.

Executive Summary

TOTAL RECORDS	TOTAL DIMENSIONS	DATA QUALITY SCORE
20	**4**	**100/100**
MISSING DATA RATE	DUPLICATE RECORDS	DETECTED OUTLIERS
0.0%	**0**	**0**

This report presents a structured analysis for **customer_metrics.csv** (Version 2). The dataset comprises **20 observations** across **4 attributes**. Empirical evaluation yields an overall Data Quality Score of **100/100**, measuring record completeness (100%), uniqueness (100%), and validity (100%).

Key Summary Findings:

• The dataset contains 20 verified observations across 4 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'tenure_months' and 'monthly_charges' (r = 0.97, strong positive).

• Categorical concentration observed in 'contract_type' where 'Two year' accounts for 60.0% of total entries.

Dataset Overview

DATASET NAME	**customer_metrics.csv**	ANALYSIS DATE	07 October 2026, 20:24 UTC
ROW COUNT	20	COLUMN COUNT	4
FILE FORMAT	CSV	MEMORY FOOTPRINT	1.7 KB

Column Overview:

Column	Data Type	Non-null	Missing	Unique	Summary
customer_id	int64	20	0 (0.0%)	20	[1.0 to 20.0], Med: 10.5
tenure_months	int64	20	0 (0.0%)	20	[1.0 to 63.0], Med: 26.5
monthly_charges	float64	20	0 (0.0%)	20	[30.0 to 112.0], Med: 80.5
contract_type	str	20	0 (0.0%)	3	Top: Two year (60%)

Data Quality

Quality Dimension	Score	Benchmark	Assessment Detail
Completeness	**100%**	>= 95.0%	Calculates proportion of non-null cells across the observation matrix.
Uniqueness	**100%**	>= 98.0%	Measures absence of identical duplicate row vectors.
Validity	**100%**	>= 90.0%	Evaluates presence of constant zero-variance features.
Consistency	**100%**	>= 85.0%	Checks for type homogeneity and uniform data formatting.

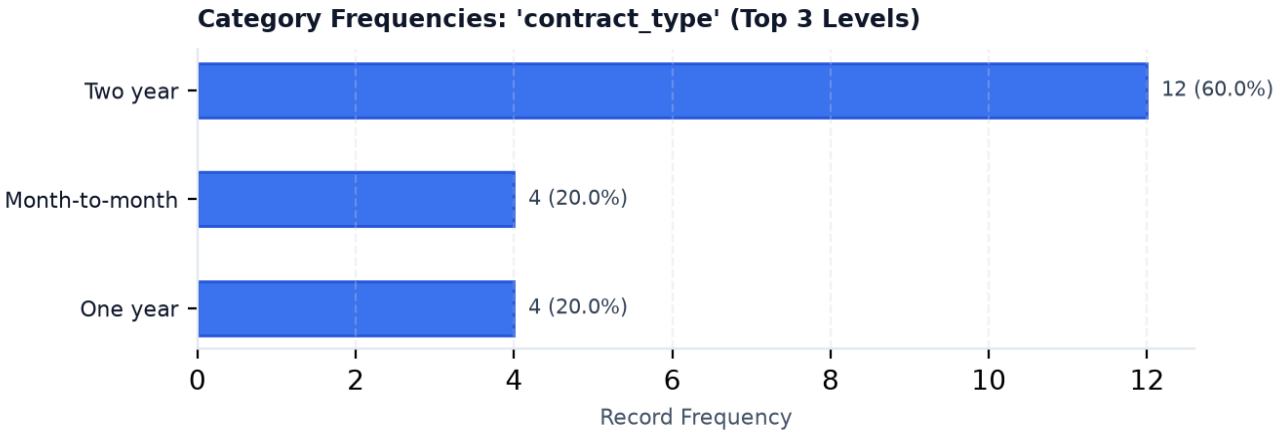
Audit Details: 0 missing cells identified (0.0% of dataset). Duplicate row analysis identified 0 duplicate record(s) (0.0%).

Descriptive Statistics

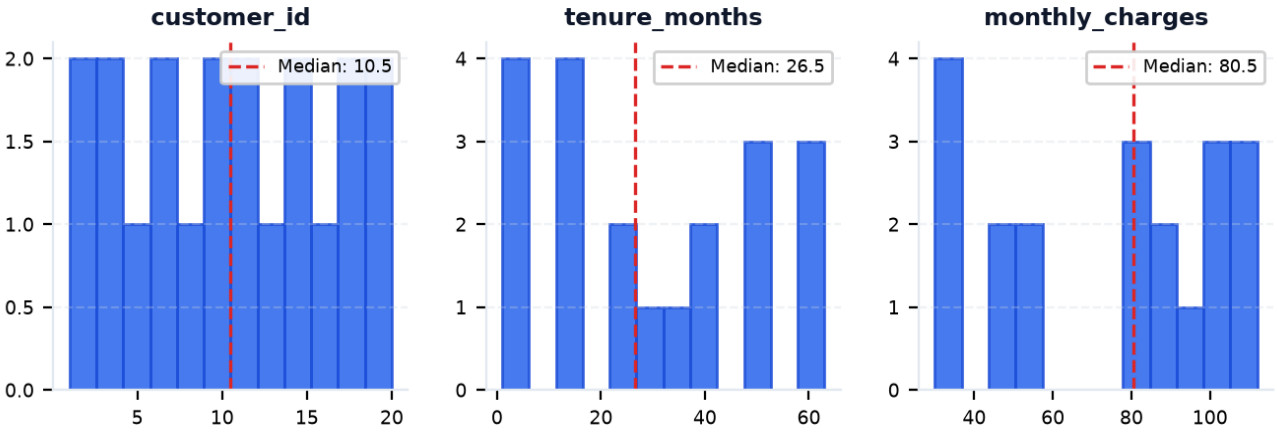
Feature	Count	Mean	Median	Std Dev	Min	Max	Q1	Q3	IQR	Skew
customer_id	20	10.5	10.5	5.9	1.0	20.0	5.8	15.2	9.5	0.00
tenure_months	20	29.6	26.5	21.1	1.0	63.0	13.5	48.5	35.0	0.20
monthly_charge	20	73.8	80.5	29.6	30.0	112.0	50.0	100.0	50.0	-0.31

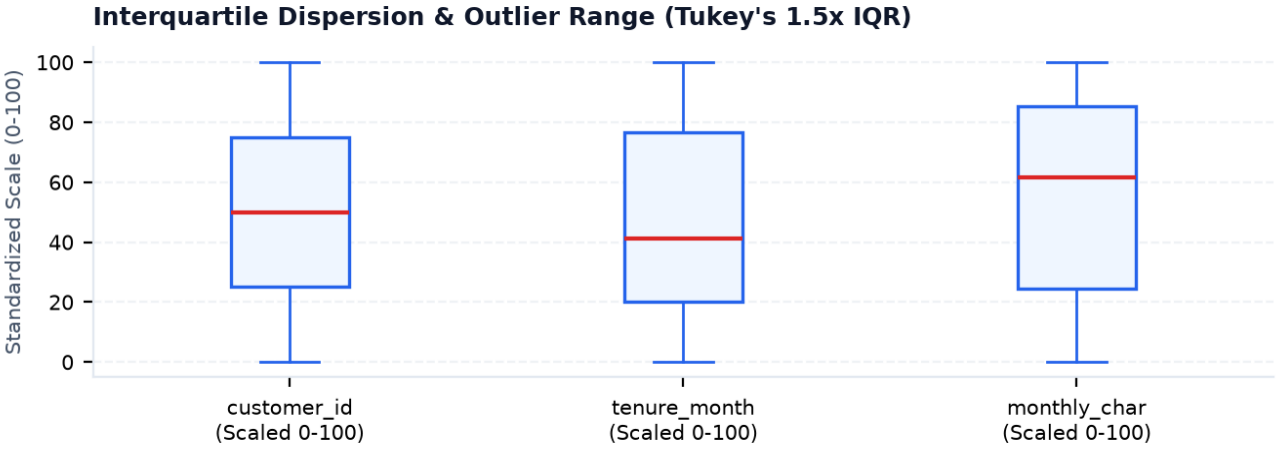
Categorical Analysis

Column	Unique	Most Frequent Value	Frequency	Percentage	Status
contract_type	3	Two year	12	60.0%	Dominant (60.0%)

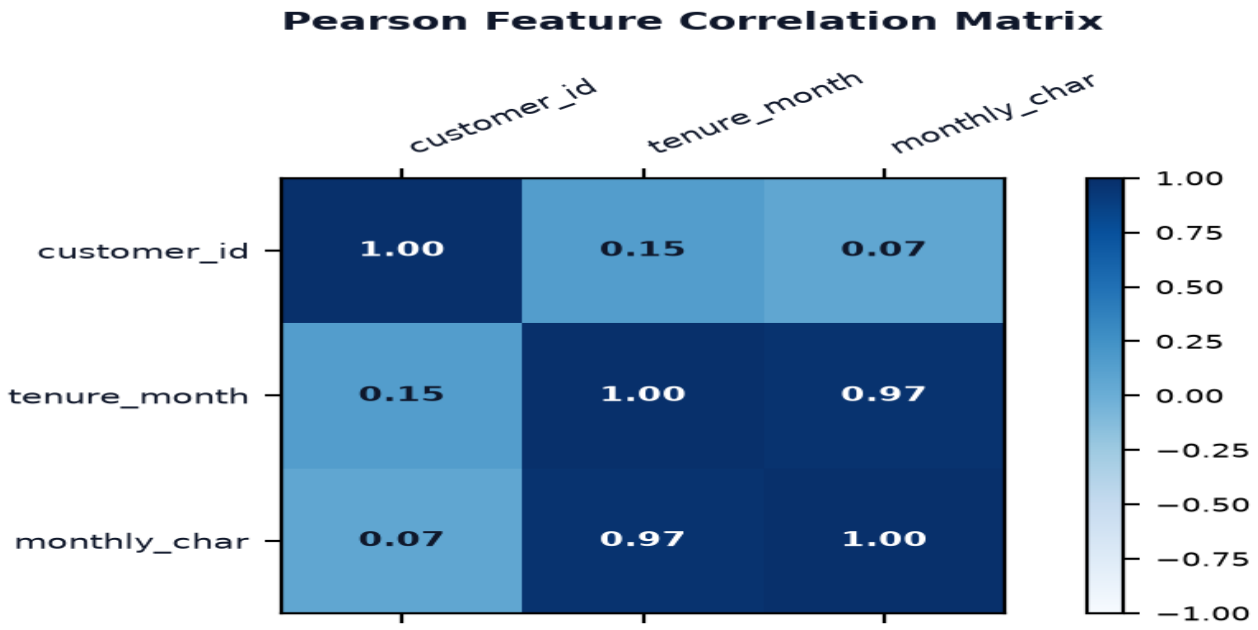


Distribution Analysis



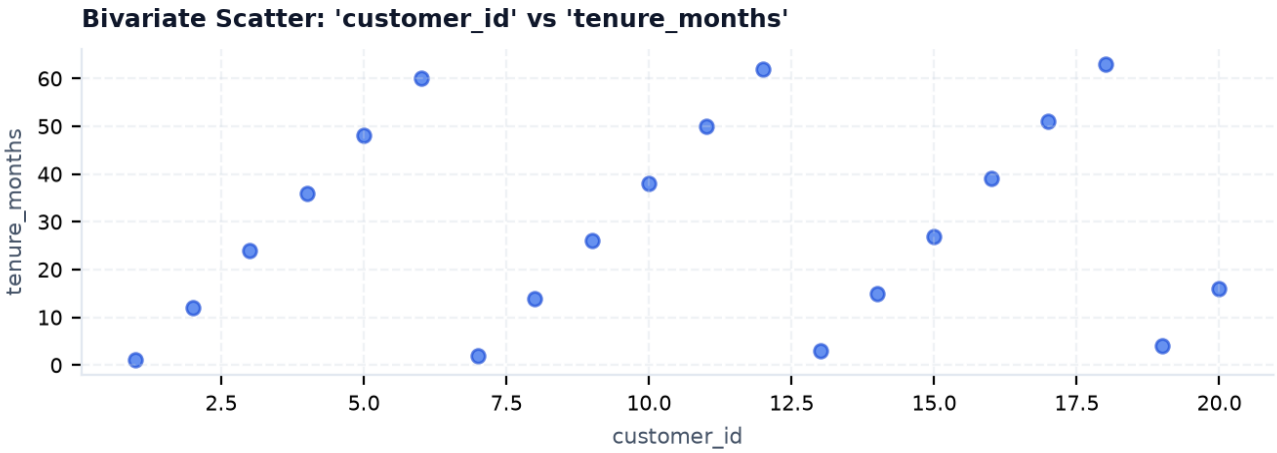


Correlation Analysis



Feature X	Feature Y	Pearson r	Strength	Direction
tenure_months	**monthly_charges**	0.97	Strong positive	Positive
customer_id	**tenure_months**	0.14	Little or no linear relationship	Positive
customer_id	**monthly_charges**	0.07	Little or no linear relationship	Positive

Relationships



Outlier Analysis

Feature	Outlier Count	Percentage	Lower Bound (1.5x IQR)	Upper Bound (1.5x IQR)
customer_id	0	0.0%	-8.50	29.50
tenure_months	0	0.0%	-39.00	101.00
monthly_charges	0	0.0%	-24.99	174.98

No statistical outliers detected outside standard 1.5x IQR thresholds.

Key Findings

• The dataset contains 20 verified observations across 4 attributes with an overall Data Quality Score of 100/100.

• Data completeness is 100%, with zero missing cells across all columns.

• Strongest linear association exists between 'tenure_months' and 'monthly_charges' (r = 0.97, strong positive).

• Categorical concentration observed in 'contract_type' where 'Two year' accounts for 60.0% of total entries.

Recommendations

Category	Priority	Action Recommendation
Operational	**Low**	Monitor the dominant category 'Two year' in 'contract_type' to identify potential sampling bias or operational dependency. Further domain validation is recommended.
Analytical	**Medium**	Evaluate 'tenure_months' and 'monthly_charges' for potential interaction terms or collinearity redundancy in multivariate regression.
Data Quality	**Low**	Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings.

Appendix

Mathematical Methodology	• **Tukey's IQR Outlier Rule:** Outliers identified strictly outside [Q1 - 1.5*IQR, Q3 + 1.5*IQR]. • **Pearson Correlation:** Linear association computed as r = cov(X, Y) / (std(X) * std(Y)). • **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) measuring distribution symmetry.
Data Quality Scoring	Overall score computed via weighted evaluation: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale directly with missing cell frequency and duplicate rows.
Compilation Provenance	Generated by Aether Engine • ReportLab PDF Architecture • Python 3.0.5 / NumPy 2.5.2. Timestamp: 07 October 2026, 20:24 UTC.